

Regional Classification — Zomato Delivery

Team Hmmmmmmmmmm · Suchit SM · Rayyan Shaikh · Ranadeep M · 18:30 deliverable · github.com/TheClazer/Zomato-delivery-estimation

GEO baseline (coords only)	Behaviour only (leak-free)	Structural (+ distance_km)	Improvement over random
1.000 AUC	0.508 AUC	0.996 AUC	GroupKFold 0.911
calibration model	≈ random — behaviour is uniform	geometry differs	survives held-out cities

The finding. Zomato's **operational behaviour** is statistically indistinguishable North vs South — all 11 univariate tests $p > 0.05$; six independent stress tests (KS, dispersion, interactions, GroupKFold, per-state, CatBoost) confirm random-level AUC for behaviour-only features. The only non-geographic feature that carries regional signal is the **haversine pickup-drop distance** — its distribution shape differs (KS D = 0.088, $p \approx 6.7 \times 10^{-10}$) and the structural model survives GroupKFold-by-city at AUC = 0.911. This isn't behaviour, it's **urban geometry**: South Indian cities have a structurally different layout of restaurants relative to customers than North Indian cities.

Six stress tests (all aligned)

#	Test	Result
1	KS shape (every numeric)	Only distance_km significant
2	Dispersion (std ratio)	All 0.98 - 1.04
3	Interaction ablation	4 of 5 add zero
4	GroupKFold by city	Behav 0.506 · Struct 0.911
5	Per-South-state classifier	0.377 vs 0.250 random
6	CatBoost cross-check	AUC 0.507

Adaptability matrix

Scenario	Effort
New city arrives	1-row edit to map
Unknown city prefix	Auto lat/lon fallback
New region definition	1-line SOUTH_STATES edit
Predict on a new CSV	python predict.py --input X
Throughput on CPU	10,130 rows / sec
Model file size	1.9 MB · loads < 100 ms

Live demo — open a terminal in the repo:

```
python predict.py --input "data/raw/Zomato Dataset.csv" --output predictions.csv
```

→ outputs prob_south for every row, AUC 0.9985 against truth, runs in 4 seconds on a laptop CPU.

Artefact	Purpose
models/region_classifier_v2_structural.txt	Trained LightGBM booster (1.9 MB, 276 trees)
models/region_classifier_v1.txt	Behaviour-only baseline booster (951 KB)
reports/regional_findings.json	Every test, every metric, every confusion matrix
reports/final_report_regional.pdf	8-page audit trail
reports/qna_elite.pdf	Q&A reference (45+ questions, all tests + models defined)
reports/pitch_regional_630pm.pdf	10-slide pitch
predict.py + notebooks/_run_regional.py	End-to-end inference + retraining scripts

If you remember only this: Behaviour 0.51 (uniform) · Structure 0.91 (geometry differs) · Geography 1.00 (clean labels). Six stress tests, two ML frameworks, all aligned. Adaptable to new cities in 1 line.